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Heuristic Function: Let us define our heuristic function $h(n)$ for A* as a pre-calculated estimated fuel expenditure from a given station's planetary system to the planet Saturn. The heuristic values are assigned as follows:

- Stations in the **Earth** system (E1–E4): $h(n) = 30$
- Stations in the **Mars** system (M1–M5): $h(n) = 20$
- Stations in the **Jupiter** system (J1–J3): $h(n) = 10$
- Stations in the **Saturn** system (S1–S4): $h(n) = 0$

More specifically each station's heuristic value can be seen in Table 1 below.

Station	E1	E2	E3	E4	M1	M2	M3	M4	M5	J1	J2	J3	S1	S2	S3	S4
Heuristic $h(n)$	30	30	30	30	20	20	20	20	20	10	10	10	0	0	0	0

Table 1: $h(n)$ values for each station

1.1 Search Q1 [1pt]

Is the heuristic $h(n)$ admissible and/or consistent in this problem? Select all that apply.

Multiple select.

- ■ The heuristic $h(n)$ is admissible and consistent.
- ☐ The heuristic $h(n)$ is admissible but NOT consistent.
- ☐ The heuristic $h(n)$ is consistent but NOT admissible.
- ☐ The heuristic $h(n)$ is neither admissible nor consistent.

1.2 Search Q2 [1pt]

Let us use the A* algorithm using the heuristic $h(n)$ to calculate the optimal delivery route from E1 to S3. Provide the path found by the A* algorithm.

Answer as a sequence of stations (fill in the blank): 51

1.3 Search Q3 [1pt]

Command wants us to calculate the optimal path to any space station. Unfortunately there is no budget to pre-calculate new heuristics between planetary systems so we have to make use of our existing heuristic $h(n)$. For which of the following goals will usage of the $h(n)$ heuristic for A* guarantee the return of the optimal path to that goal? Note that the starting station is still E1. Select all that apply.

Multiple select.

- ☐ E2 E1 -> E2 : cost 2, h(E2) = 30 overs estimates a lot
- ☐ E3 E 1 . E2 > E3 , cost = 2+3 = 5, h(E3) = 30
- ☐ E4 same as above
- ☐ M1 2+3+10+2 +3+4 = 24, h(M1) = 20
- ☐ M2

(4.2) $f(n) = g(n) + h(n)$ E1 → S3 (goal)

$g(E1) = 0$, $h(E1) = 30$, $f(E1) = 0 + 30 = 30$

E1 $\xrightarrow{2}$ E2 $\xrightarrow{3}$ E3 $\xrightarrow{10}$ M3 $\xrightarrow{2}$ M4 $\xrightarrow{3}$ M5 → J2 → J3 → S4 → S1 → S2 → S3

(A*)

E1 → E2 : 2
E2 → E3 : 3
E3 → M3 : 10
M3 → M4 : 2
M4 → M5 : 3
M5 → J2 → 10
J2 → J3 : 4
J3 → S4 → 10
S4 → S1 → 3
S1 → S2 : 2
S2 → S3 : 2

cost → 51

(1.3) Start ⇒ E1 $h(n) = 30$

goal.

E2 : E1 → E2 : 2 (30) overestimates

E3 E1 → E2 → E3 : 2+3 = 5 overestimate

E4 2+3+4 = 9, $h(E4) = 30$ overestimate

M1 E1 → E2 → E3 → M3 → M4 → M5 → M1 = 24
$h(M1) = 20$ underestimate

M2 2+3+10+2+3+4+1 = 25 underestimate

M5 E1 → E2 → E3 → M3 → M4 → M5 : 2+3+10+2+3 = 20
$h(M5) = 20$, Admissible & Consistent

J1 – 2+3+10+2+3+10+4 = 39
$h(J1) = 10$, underestimate.

S1 (47), $h(S1) = 0$. underestimate

S2 (49), $h(S2) = 0$, underestimate

S4

E1 → E2 → E3 → M3 → S4

2+3+10+30 = 45

$h(S4) = 0$ underestimate.

1.4 $h(n)$ for J increased by +10

⇒ $f(n) = g(n) + h(n)$

$f(n)$ will increase (+10) ✓

⇒ might change path

⇒ no change – path cost

⇒ order of node exploration (change)

- [x] M5

cost = 2+3+10+2+3 = 20, h(M5) = 20

- [ ] J1
- [ ] S1
- [ ] S2
- [ ] S4

1.4 Search Q4 [2pts]

Suppose a hacker secretly changes the heuristic function $h(n)$ for Stations J1, J2, and J3 by increasing their heuristic values by +10. How does this adjustment impact the A* search and the path found by A*? Select all that apply.

Multiple select.

- [x] ~~The $f(n)$ values at J1, J2, and J3 will remain the same.~~
- [x] The overall path found by A* will remain the same.
- [x] The cost of the overall path found by A* will increase.
- [x] The heuristic adjustment will change the order in which nodes are explored in the A* search.

h(J) = 10+10 = 20

$f(n) = g(n) + h(n)$

No path cost impact - order may change.

Optimization

magic square - (sum to same val) row, col, diagon

(1) no. of valid square (3x3) - [1, 2, ... 9]

$= 9! = 362,880$

valid = (8) (both) total 8 solution for 3x3 magic sq.

$P(\text{magic sq}) = \frac{8}{362,880}$

Exp attempt $= 1/P(\text{magic sq}) = \frac{362,880}{8} = 45360$

(2)

1	3	8
2	4	7
9	6	5

target sum $= -15$

$|1+3+8-15| = 3$

$|2+4+7-15| = 2$

$|9+6+5-15| = 5$

7	9	5
3	6	2
8	1	4

Row
$|7+9+5-15| = 6$
$|3+6+2-15| = 4$
$|8+1+4-15| = 2$

Colm
$|7+3+8-15| = 3$
$|9+6+1-15| = 1$
$|5+2+4-15| = 4$

Diagn
$|7+6+4-15| = 2$
$|8+6+5-15| = 4$

$= 26$

(23)

6, 5 4
7, 8, 1
9, 3, 2

⇒

6	4	5
7	8	1
9	3	2

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \begin{matrix} = 16-15| = 9 \\ = |15-15| = 0 \\ = |24-15| = 9 \end{matrix}$$

$1+4+7-15 = 3$
$2+5+8-15 = 0$
$3+6+9-15 = 3$

$1+5+9-15 = 0$
$3+5+7-15 = 0$

Fitness $= 9+0+9+3+0+3+0+0 = 24$

1 swap

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 9 & 6 \\ 7 & 8 & 5 \end{bmatrix} \quad \begin{matrix} 9 \\ |19-15| = 4 \\ |20-15| = 5 \end{matrix}$$

col-
$|12-15| = 3$
$|19-15| = 4$
$|14-15| = 1$

diag - $|19-15| = 4$
$|15-15| = 0$

$\Rightarrow 9+4+5+8+4+0$
$= 30$

(3)

$$\begin{bmatrix} 4 & 2 & 3 \\ 1 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \begin{matrix} 0 \\ 6 \\ 3 \\ 9 \end{matrix}$$

$3 \quad 0 \quad 3$

Fitness $= 18+6+3 = 27$

$$0\begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix} \begin{matrix} 3 \\ 9 \\ 3 \\ 6 \end{matrix}$$

$3 \quad 0 \quad 3$

fitness = (24)

(2.6)

P1 | A | B | C | D | E | F |

P2 | G | H | I | J | K | L |

2pt Crossover

Child 1 | A | B | I | J | E | F |

C2 | G | H | C | D | K | L |

Uniform Crossover —

Child 1 | A | H | C | J | E | L |

Child 2 | G | B | I | D | K | F |

Ⓐ 1st & 3rd no.

P1 = [ABC | DEF]

P2 = [GHI | JKL]

C1 = [ABC | JKL]

C2 = [GHI | DEF]

May be Repetitive

Pop. size = 10

Uniform Crossover → may lead to Duplicate / Invalid

③ Uniform Crossover with Duplicating check ✓

④ P1 [ABC | DEF]
P2 = [GHI | JKL] > C1 = [ABC | DEF]

unique no. added ✓

⇒ no duplication / repeat

2.3 Optimization Q3 [1pt]

An important step in Genetic Algorithms involves mutations. We define mutations in this problem as a random swapping of two digits in a valid square. No more than one swap can occur per mutation. An example of a mutation can be seen in Figure 6 below.

6	5	4
7	8	1
9	3	2

(a) Before mutation

2	2	8
6	1	9
5	3	7

(b) After mutation

Figure 6: Mutation where the digits 5 and 4 are swapped.

Given the valid square in Figure 7, what is the **best** fitness (smallest fitness score value) that can be obtained from the square with one mutation?

1	2	3
4	5	6
7	8	9

Figure 7: Square for Q3 and Q4

Answer as a whole number. 12

2.4 Optimization Q4 [1pt]

Given the valid square in Figure 7, what is the **worst** fitness (largest fitness score value) that can be obtained from the square with one mutation?

Answer as a whole number. 30

2.5 Optimization Q5 [1pt]

Our Genetic Algorithm follows the following procedure:

1. Generate a valid square at random
   (a) If the valid square is a magic square solution, stop and return the square.
   (b) Otherwise, the valid square is NOT a magic square solution – generate a new valid square by randomly swapping two elements of the current square (a mutation).
      i. If the fitness score of the new square is better than the current square, set the current square = new square.
      ii. Otherwise the current square does not change.
2. Perform as many iterations as needed until a magic square solution is found.

Assume we want to improve our algorithm and include crossover in addition to mutation. Select all of the **valid** crossover schemes below. Assume a constant population size of 10 where parents are chosen proportional to their fitness and every child has only two parents. Select all that apply.

Multiple select.

- ☐ Two-point crossover. The first and third row of one parent is combined with the second row of the other parent.
- ☐ Uniform crossover. A child is created by alternately choosing from each parent. The new square will be composed of 5 values from one parent and 4 from the other.
- ☐ The children are produced by a uniform crossover that works from left to right and top to bottom. If a value already exists in the child (e.g. a 4 is chosen from one parent to pass onto the child but it is already present in the child from another parent) the value at that same position from the other parent is used instead.
- ☐ A child has the first two rows of one parent. Then, the third row of the other parent is added one by one to the third row of the child. If any of the values in the third row also appear somewhere in the first two rows of the child, a number that hasn't appeared in the child yet is used instead.

Let us define the evaluation function for determining the utility value of a game state as the maximum possible moves available to a player in a given game state. For example, from the perspective of Player 1, an empty board has a utility value of $4 \cdot 16 = 64$ since each type of piece has 16 possible positions for placement.

3.1 Game Playing Q1 [1pt]

What is the utility value of the game state for Player 2 after Player 1 has placed down one piece?

Answer as a whole number. 53 8+15+15+15

3.2 Game Playing Q2 [1pt]

Notice that from an empty board, there are 64 possible game states that can result from Player 1's first move. In order to reduce the search space and speed up our game playing algorithm, we can take advantage of symmetry properties of Quantik game states to reduce equivalent game states by symmetry. If we take into account **rotational symmetry about the center of the board, reflective symmetry across the vertical/horizontal/diagonal axes, and substitution symmetry (where piece types can be substituted for each other as shown in Figure 14)**, how many unique game states are there after Player 1 makes the first move on an empty board?



(a) State X (b) State Y

Figure 14: State X and State Y are equivalent by substitution symmetry

Answer as a whole number. 64/4 * 1/2 * 1/4 = 2

Total - (4). ↳ reflection

Substitution symmetry = $\frac{N}{4}$

Rotational = 64/4

Player 1 is trying to decide what their best first move is for Quantik. They decide to make a Game Tree of depth 3 to help evaluate their options (for simplicity they have ignored some nodes). The game tree is shown in Figure 15 below, use this game tree for Q3-Q6.

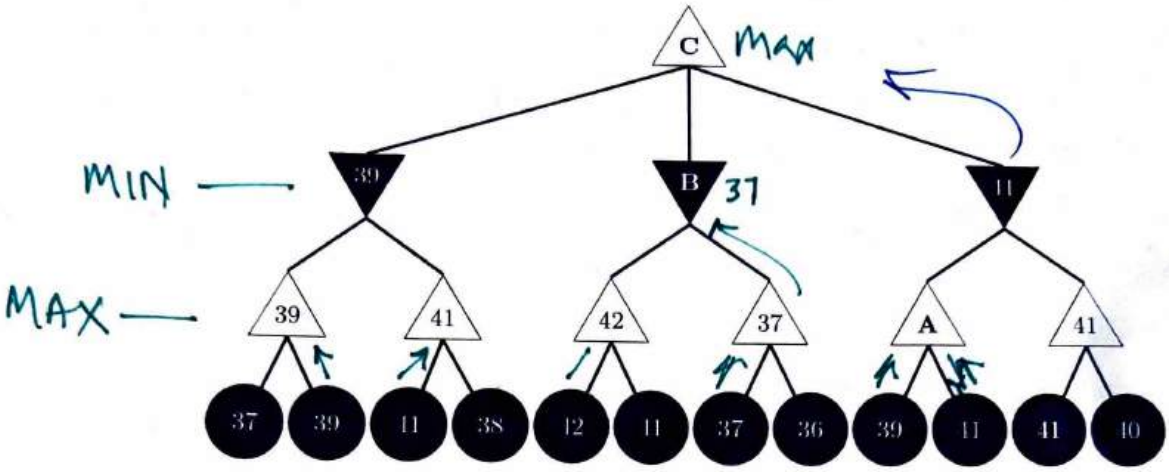


Figure 15: Game Tree

3.3 Game Playing Q3 [1pt]

Using minimax to propagate utility values up the tree, what is the utility value of **node A**?

Answer as a whole number. 41

3.4 Game Playing Q4 [1pt]

Using minimax to propagate utility values up the tree, what is the utility value of **node B**?

Answer as a whole number. 37

3.5 Game Playing Q5 [1pt]

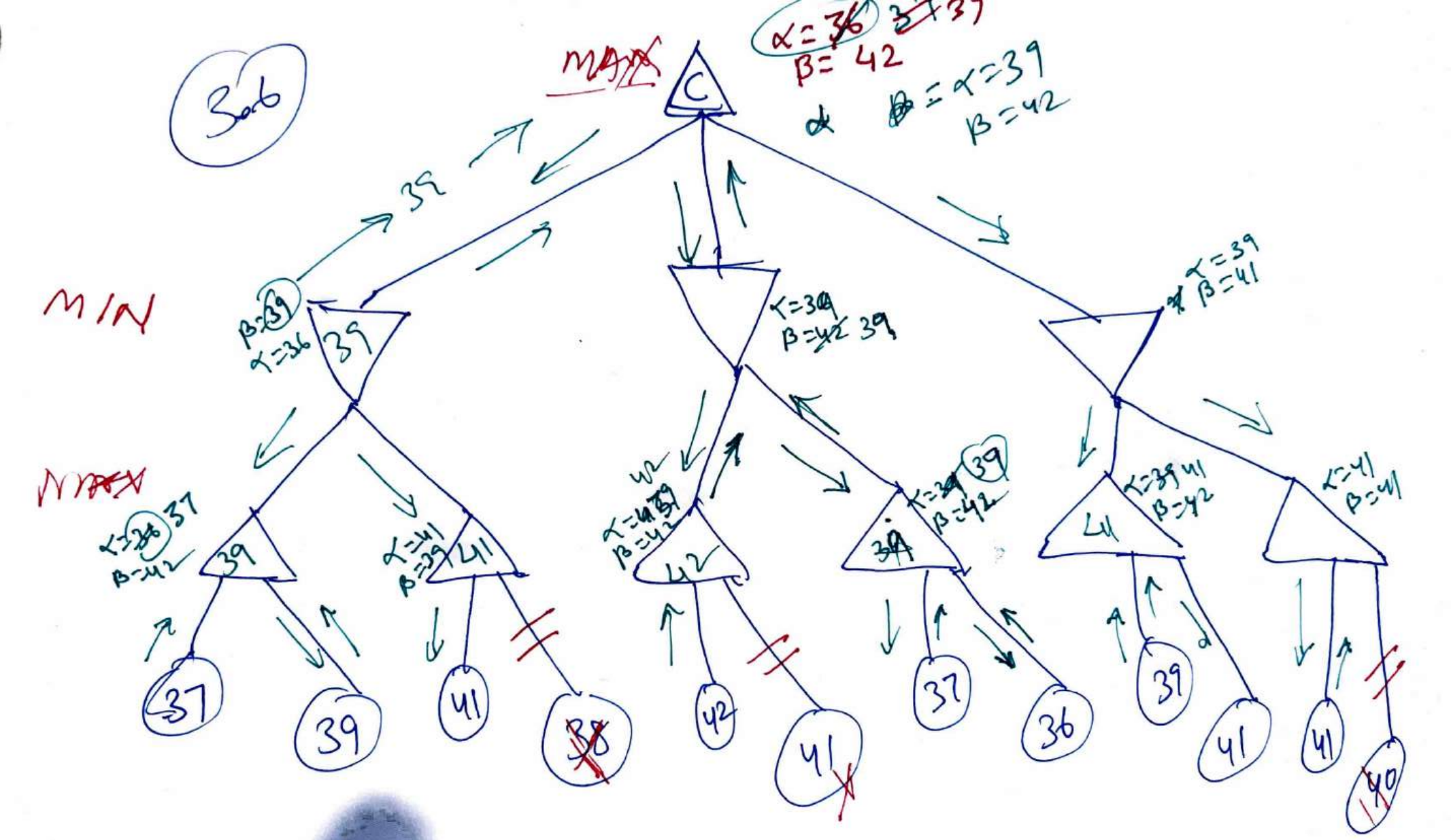
Using minimax to propagate utility values up the tree, what is the utility value of **node C**?

Answer as a whole number. 41

3.6 Game Playing Q6 [1pt]

How many leaf nodes will be pruned if we apply Alpha-Beta pruning from left to right to the Game Tree in Figure 15? Assume that the maximum possible utility is 42 and the minimum possible utility is 36.

Answer as a whole number. 3


Soln
MAX
C
α = 36 → 39
β = 42
α = 39
β = 42
MIN
β = 39
α = 36
39
α = 39
β = 42
39
α = 39
β = 41
MAX
α = 36 37
β = 42
39
α = 41
β = 39
41
42
α = 39 39
β = 42
39
α = 39 41
β = 42
41
α = 41
β = 41
37
39
41
38
42
41
37
36
39
41
41
40

3.7 Game Playing Q7 [1pt]

In general, how does minimax compare to alpha-beta pruning in space and time complexity? Select all that apply.

Multiple select.

- [x] Alpha-beta pruning's time complexity is always better than or equivalent to minimax's
- [ ] Alpha-beta pruning always has a higher space-complexity since it needs to keep track of the alphas and betas.
- [ ] Minimax's time complexity depends linearly on the depth
- [x] Minimax's time complexity depends exponentially on the branching factor $O(b^d)$
- [ ] For smaller, shallower game trees, minimax is faster than alpha-beta due to lower overhead.

maybe

Alpha-Beta Time Complexity $O(b^d)$ — Minimax

CSP

Variables – S, M, L

Domain – A, B, C, D, E, F, G, H (8)

11 – complete and consistent assignments.

① total complete = $(\text{Domain})^n = (8)^3 = 512$

4.2 forward checking.

State 0 at C ←

S : {A, B, C, D, E, F, G, H}

M : {A, B, C, D, E, F, G, H}

L : {A, B, C, E, G, H}

Find check S — Allowed —

(M, S) : (A,A), (A,B), (A,C)

<C, S> (B,A), (B,B), (B,C)

(C,A), (C,B), (C,C) valid.

(D,E)

(E,D), (E,E)

(F,G), (G,F), (G,G), (H,H)

S = {A, B, C} valid domain for S

Unary constraint L≠D, L≠F

Check L

<L, M> = <L, C>

valid ⇒ (A,C), (B,C), (D,C), (E,C), (F,C)

L = {A, B, D, E}

final S: {A, B, C}, M = C, L {A, B, D, E} ⇒ L = {A, B}

5 removed, 3 removed = 8

⇒ 5 + 3 = 8

Binary Constraints

We've reduced the binary constraints into allowed pairs format between each variable for your ease of use.

Edge	Total	Allowed Tuples
(Large, Medium)	35	(A, A), (A, B), (A, C), (A, D), (A, E) (B, A), (B, C), (B, D), (B, E) (C, A), (C, B), (C, D), (C, E) (D, A), (D, B), (D, C), (D, E), (D, F), (D, G) (E, A), (E, B), (E, C), (E, D), (E, F), (E, G) (F, D), (F, E), (F, G), (F, H) (G, D), (G, E), (G, F), (G, H) (H, F), (H, G)
(Medium, Small)	16	(A, A), (A, B), (A, C) (B, A), (B, B), (B, C) (C, A), (C, B), (C, C) (D, E) (E, D), (E, E) (F, G) (G, F), (G, G) (H, H)
(Large, Small)	17	(A, A), (A, F), (A, G) (B, B), (B, H) (C, D), (C, E) (D, C), (D, E) (E, C), (E, D) (F, A), (F, G) (G, A), (G, F), (G, G) (H, B)

Table 4: The Binary Constraint Table

For those of you that are interested, this problem has 11 complete and consistent assignments. Alright let's get to work and make sure everyone receives their Christmas presents on time!

4.1 CSP Q1 [1pt]

Determining what to define as the variables and what to define as the domains for CSP problems is extremely important. If we had defined the ports as the variables in our approach to this CSP problem, we'd have had 8 variables with a search space of 16777216 complete assignments! By defining the ships as the variables, our search space is significantly smaller.

How many **complete assignments** are there in total for this CSP problem? For an assignment to be complete, all variables must be assigned a single value from its starting domain and the assignment does not have to be consistent. (*AIMA 4th Ed. Section 6.1*)

Answer as a whole number.

(4.3) Domain State one (S): {A, B, C, D, E, F, G, H}
M: {A, B, C, D, E, F, G, H}
(L): {A, B, C, E, G, H}

– path consistency –

A
B
C
D
E
G
H

$\{x_i, x_j\}$ L: {L=x, S=y}

$\{x_i = a, x_j = b\}$ M=z for which

(L=x, M=z) valid in (L,M

(M=z, S=y) valid in (M,

– (L=A) (L,M): ~~(A,A)~~, ~~(A,B)~~, (A,C), ~~(A,D)~~, ~~(A,E)~~

~~(M,S): (A,A), (A,B), (A,E)~~

~~M = A, B, C~~

(L=A, M) → valid (A,A) (A,B) (A,C) (A,D) (A,E)

possible M (M,S) ? (M=A) (A,A) (A,B) (A,C)

M=B (B,A) (B,B) (B,C)

(C,A) (C,B) (C,C)

(M=E) (E,D)

for $\phi(L,S)$ → valid assign M

→ satisfy binary

b/w (L

$\rightarrow (M) \rightarrow S$

4.3 CSP Q3 [2pts]

Given the node and arc consistent Domain State One in Figure 18, make **Large** and **Small** path consistent with respect to **Medium**. What values are removed from the domain of **Large** as a result? Select all that apply.

Multiple select.

- ☐ A
- ☐ B
- ☐ C
- ☐ E
- ☐ G
- ☐ H

4.4 CSP Q4 [2pts]

Given the node and arc consistent Domain State One in Figure 18, make **Large** and **Small** path consistent with respect to **Medium**. What values are removed from the domain of **Small** as a result? Select all that apply.

Multiple select.

- ☑ A
- ☑ B
- ☑ C
- ☐ D
- ☐ E
- ☐ F
- ☑ G
- ☑ H

5 Probability [5pts]

Introduction

Probability and combinatorics are closely related. Combinatorics studies countable sets, while probability uses combinatorics to assign probabilities (values between 0 and 1) to events. Real-world machine learning tasks frequently involve combinatorial structures as we focus on how to model, infer, or predict using graphs, matchings, hierarchies, or other discrete structures behind the data. Moreover, "thinking probabilistically" is a good approach to navigating complex decisions in life, as we want to make decisions that "maximize our expected value".

5.1 Probability Q1 [1pt]

Let X and Y be i.i.d. variables following the normal distribution $\sim \mathcal{N}(0,1)$. What is the probability that Y is greater than $10X$? Compute the value of $P(Y > 10X)$.

Hint: You do not need to round. 0.5

Answer as an exact decimal (no rounding).

5.2 Probability Q2 [1pt]

Suppose that two integers a and b are uniformly randomly selected from the set $S = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ with replacement. Find the probability that $\max(0, a) = \min(0, b)$.

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 56.*

Answer as a whole number. 25/81 = p+q = 25+81 = 106

5.3 Probability Q3 [1pt]

Your friend proposes a game as follows: in each round, a wheel uniformly numbered 1 through 6 is spun four times. If any of the four numbers match, Player 2 gives Player 1 a dollar. If none of the four numbers match, Player 1 gives a dollar to Player 2. What is the expected value for the number of dollars Player 1 will earn in one round?

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 5.*

Answer as a whole number. 4/9 , 4+9 = 13

Probab

① Sec 19.4

X, Y – indep & identically distributed.

$P(X_j) = P(X_{j+1}) = P(X_{j+2}) \cdots$

$P(X_j) = P(X_j \mid X_{j-1}, X_{j-2} \cdots)$

Normal Dist $\sim N(0,1)$ ⇒ Mean $= 0$, Variance $= 1$ ⇒ $E[X] = 0$, $E[Y] = 0$

$P(Y > 10X)$??

$\Rightarrow P(Y - 10X) > 0$

Let $Z = Y - 10X \Rightarrow P(Z) > 0$

$\text{variance}(Z) = \text{var}(Y) + 10^2 \times (\text{var}(X))$

$\Rightarrow \text{var}(Z) = 1 + 100 = 101$

$Z \sim N(0, 101)$

$P(Z) > 0 = P(Z) < 0$

$\Rightarrow \left(\frac{1}{2}\right) = 0.5$

② $S = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ total options $= 9 \times 9 = 81$

(a, b) random select

$\max(0, a)$ if $a \le 0 \Rightarrow 0$

if $a > 0 \Rightarrow a$

$\min(0, b)$, if $b \ge 0 \Rightarrow 0$

$b < 0 \Rightarrow b$

$\max(0,a) = \min(0,b)$

if $= 0$ ✓

or $\max(0,a) = \min(0,b) = x > 0$ ~~X can't be > 0~~

a ~~can be~~ $\{-4, -3, -2, -1, 0\}$ $b: \{-4, -3, -2, -1, 0\}$

valid $a, b = 5 \times 5 = 25$

$\frac{25}{81}$

$81 - 25 = 56$ ✓

(3) Total outcome = $6 \times 4 = 24$

4 spins

P(all distinct)

$$E(X) = \sum_i x_i P(X = x_i)$$

1st — 6
2nd — 5
3rd — 4
4th — 3
options.

fav outcome = $6 \times 5 \times 4 \times 3 = 360$

$P(\text{all uniq}) = \frac{360}{1296} = \frac{5}{18}$

$P(\text{min 1 match}) = 1 - P(\text{all uniq}) = 1 - \frac{5}{18} = \frac{13}{18}$

P1 wins = $\frac{13}{18}$, P1 loses = $5/18$

$E = 1 \cdot \frac{13}{18} + (-1) \times \frac{5}{18} = \frac{8}{18} = \frac{4}{9}$

5.4 Probability Q4 [1pt]

Your friend is back with a new game! In each round, a wheel uniformly numbered 1 through 6 is spun repeatedly until the number 4 appears. You win an amount in dollars equal to the highest number that is encountered during the spins. What is the expected value for the amount of dollars you will win from one round of the game?

Hint: Think about the order in which the numbers 4, 5, and 6 appear during the spins. What are the probabilities of each number being the highest before the round ends? Consider how permutations of these numbers affect the outcome.

Hint: The answer is a whole number.

Answer as a whole number.

5.5 Probability Q5 [1pt]

Consider a bag of marbles containing red and green marbles where the probability of drawing a red marble from the bag is 0.5 and the probability of drawing a green marble from the bag is also 0.5. Two friends, A and B, each have their own bag of marbles and they continuously draw marbles from their own bags with replacement, stopping when they draw a red marble. What is the probability that person A draws more marbles than person B?

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 2.*

Hint: Consider the probability distribution of the number of marbles each friend draws until they get a red marble. Using symmetry in the problem can simplify the calculation.

Answer as a whole number.

(5) $P(g) = P(r) = 0.5$ $\quad (P(N_A > N_B)\ ?)$

$P(N = k) = (1-p)$

$P(g)$, k-1 times $= (0.5)^{k-1}$

$P(\text{red @ } k) = (0.5)^{k-1} \times 0.5$

$P(X = x) = p \cdot q^{(x-1)}$ ⟹ geometric distribution

$P(X \le x) = 1 - (1-p)^x$

$P(N_A > N_B) = P(N_B > N_A)$

$P(N_A > N_B) + P(N_B > N_A) + P(N_A = N_B) = 1$

$2P(N_A > N_B) + P(N_A = N_B) = 1$

$P(N_A = k) = (0.5)^{k-1} \times 0.5$

$P(N_A = N_B) = (0.5)^{k-1} \times 0.5) \times ((0.5)^{k-1} \times 0.5)$

$= (0.5)^{2k-2} \times 0.25$

$P(N_A = N_B) = \sum_{n=1}^{\infty} (0.5)^n \times (0.5)^n = \sum (0.5)^{2n}$

~~$= 4(P(4)) + 5(P(5)) + 6(P(6))$~~

$= \frac{4 \times 1}{5} + \frac{5 \times 1}{7} + \frac{6 \times 1}{9} = \frac{15}{8} = 5$

$P(N_A = N_B) = \frac{q}{1-r} = \frac{0.25}{1-0.25} = \frac{0.25}{0.75} = \frac{1}{3}$

$P(A > B) + P(B > A) + P(A = B) = 1$

$P(A > B) + P(A < B) + \frac{1}{3} = 1$

$2P(A > B) = 1 - \frac{1}{3} = \frac{2}{3}$

$P(A > B) = \frac{1}{3}$ ⊛

Cell Proliferation (P)	Apoptosis (A)	DNA Repair (R)	Cancer (C)	$p(C\|P, A, R)$
Normal	Not Triggered	Effective	Absent	0.8
Normal	Not Triggered	Effective	Present	0.2
Normal	Not Triggered	Ineffective	Absent	0.6
Normal	Not Triggered	Ineffective	Present	0.4
Normal	Triggered	Effective	Absent	0.4
Normal	Triggered	Effective	Present	0.6
Normal	Triggered	Ineffective	Absent	0.7
Normal	Triggered	Ineffective	Present	0.3
Increased	Not Triggered	Effective	Absent	0.6
Increased	Not Triggered	Effective	Present	0.4
Increased	Not Triggered	Ineffective	Absent	0.8
Increased	Not Triggered	Ineffective	Present	0.2
Increased	Triggered	Effective	Absent	0.3
Increased	Triggered	Effective	Present	0.7
Increased	Triggered	Ineffective	Absent	0.5
Increased	Triggered	Ineffective	Present	0.5

0.2 = 2/10 = 1/5

6.1 Bayes Nets Q1 [1pt]

P(C=Present | D=Low, M=Pres, P=Increas, R=Ineff, A=NT, S=Pres)

What is the probability of a cell being a cancer cell (C = Present) given that there is no DNA damage (D = Low), there is a mutation in the p53 gene (M = Present), there is a lot of cell proliferation (P = Increased), DNA repair is ineffective (R = Ineffective), apoptosis has not been triggered (A = Not Triggered), and the cell is having stress signals (S = Present)?

1/6

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 4.*

Answer as a whole number. 6

6.2 Bayes Nets Q2 [1pt]

In the question above (Q1), C is D-separated from which of the following variables? Select all that apply.

Multiple select.

- ☐ D
- ☐ S
- ☐ M
- ☐ R
- ☐ A
- ☐ P

No direct path b/w D/S

D → M ← S

M → R, M → P, M → A

R, P, A → C

M → R → C
M → A → C
M → P → C

× direct impact

6.3 Bayes Nets Q3 [2pts]

What is the probability of the cell having a high DNA damage (D = High), given no stress signals (S = Absent) and the mutation of the p53 gene is absent (M = Absent)? For this question only, assume $P(S|D) = P(S)$.

P(D=High | S=Absent, M=Absent)

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 21.*

Answer as a whole number.

= P(S=Ab, M=Abs | D=High) × P(D=High)
× P(S=Ab, M=Abs)

10/3 ?

=41

=

(5) $P(q) = P(r) = 0.5$

$$= \frac{P(S=Abn, M=Abn \mid D=High)\ (P(D=High))}{P(S=Ab, M=Abn)}$$

$P(S=Abn, M=Abn \mid D=High)$

$= P(M=Abn \mid D=High, S=Abn) \times P(S=Abn \mid D=High)$

$= 0.5 \times P(S=Abn \mid D=High)\ P(S)$

$= 0.5 \times 0.7 = 0.35$

Num = 0.35×0.4

Denom $P(S=Abn, M=Abn)$

$= P(S=Abn, M=Abn \mid D=High)\ (P(D=High)$

$+ P(S=Abn, M=Abn, D=Low)\ P(D=Low)$

$= 0.35 \times 0.7 + 0.7 \times 0.7 =$ nim

$= 0.7 \times 0.7 \times 0.6 + 0.35 \times 0.4$

$= 0.434$

$$\frac{0.35 \times 0.4}{0.434} = \frac{0.14}{0.434} = 0.322580645$$

$$= \frac{0.14 \times 10000}{0.434 \times 10000} = \frac{1400}{4340} = \frac{70}{217} = \frac{10}{31}$$

$P+q = 41$

6.4 Bayes Nets Q4 [3pts]

What is the probability of a cell being cancerous (C) given that the DNA repair is effective (R = Effective, that no apoptosis is occurring (A = Not Triggered), and that there's no mutation on the p53 gene (M = Absent)?

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 18.*

Answer as a whole number.

6.5 Bayes Nets Q5 [3pts]

What is the probability of the cell proliferation being increased (P = Increased), given that the apoptosis has been triggered (A = Increased), and that the DNA repair is ineffective (R = Ineffective)?

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 1167.*

Answer as a whole number.

6.6 Bayes Nets Q6 - Extra Credit [0.5pts]

What is the probability of the cell proliferation being increased (P = Increased), given that the apoptosis has been triggered (A = Triggered), that there is a mutation on the p53 gene (M = Present), and that the DNA repair is ineffective (R = Ineffective)?

The answer can be expressed in the form of p/q where p and q are relatively prime positive integers. Submit your answer as the value of $p + q$. *Hint: The absolute value of $p - q$ is 3.*

Answer as a whole number.

6.7 Bayes Nets Q7 - Extra Credit [0.5pts]

The problems above (Q5 and Q6) generate two very different solutions and inherently serve as an illustrative example of one type of D-separation. Which type of D-separation is that?

Select one.

- ☐ Chain structure ($X \to Y \to Z$; $X \perp Z|Y$)
- ☐ Fork structure ($X \leftarrow Y \to Z$; $X \perp Z|Y$)
- ☐ Collider structure ($X \to Y \leftarrow Z$; $X \not\perp Z|Y$)

(5) $P(g) = P(r) = 0.5$

(6.4) R=eff, A=NT, M=Abs

$P(C = \text{Present} \mid R = \text{eff}, A = \text{Not Trigg}, M = \text{Absent})$

$P(C = P_m \mid P = \text{Normal}, A = NT, R = Eff) = 0.2$

$P(C = P \mid P = \text{Inc}, A = NT, R = Eff) = 0.4$

$$\sum_P P(C \mid P, A = NT, R = Eff) \times P(P \mid M = \text{Absent})$$

$= P(P = \text{normal}) \times P(C = P \mid P = \text{normal}, A = NT, M = \text{Abs})$
$+ P(P = \text{increased}) \times P(C = P \mid P = \text{incr}, A = NT, M = \text{Abs})$

$0.6 \times 0.2 + 0.4 \times 0.4$

$= 0.12 + 0.16 = 0.28$

when M = Abs:
$P(P = \text{nor}) = 0.6$
$P(P = \text{incr}) = 0.4$

$\frac{704}{25} = 7 + 25 = 32$

(5) $P(P = \text{incr} \mid A = \text{Trigg}, R = \text{Ineff})$

$= \sum_M$

7.3 Machine Learning Q3 [1pt]

Which component should the data point $(15, 40)$ be assigned to?

Hint: You can use approximate values for exponentials to simplify calculations and ensure responsibilities are easy to interpret.

Select one.

- ☐ Component 1
- ☐ Component 2

Bayesian Information Criterion (BIC)

Given a dataset with $n = 10000$ data points and various Gaussian Mixture Model (GMM) configurations. The BIC (Bayesian Information Criterion) is calculated using the formula:

$$\text{BIC} = p \cdot \ln(n) - 2\ell$$

Where p is the number of parameters in the model, ℓ is the log-likelihood, and n is the number of data points. Use the natural logarithm (ln, base e). The following table provides the log-likelihoods (ℓ) and the number of parameters (p) for each model with different numbers of components k.

Model	Number of Components (k)	Log-Likelihood (ℓ)	Number of Parameters (p)
A	1	-12000	3
B	2	-11500	6
C	3	-10800	12
D	4	-9500	20
E	5	-9400	35

Table 5: Models

7.4 Machine Learning Q4 [1pt]

What is the BIC for **model A**? Round your answer to 2 decimal places.

Answer as a decimal rounded to 2 decimal places.

7.5 Machine Learning Q5 [1pt]

What is the BIC for **model B**? Round your answer to 2 decimal places.

Answer as a decimal rounded to 2 decimal places.

7.6 Machine Learning Q6 [1pt]

What is the BIC for **model C**? Round your answer to 2 decimal places.

Answer as a decimal rounded to 2 decimal places.

7.7 Machine Learning Q7 [1pt]

What is the BIC for **model D**? Round your answer to 2 decimal places.

Answer as a decimal rounded to 2 decimal places.

7.8 Machine Learning Q8 [1pt]

What is the BIC for **model E**? Round your answer to 2 decimal places. 19122.36

Answer as a decimal rounded to 2 decimal places.

7.9 Machine Learning Q9 [1pt]

Based on Q4-Q8 what is the model that best fits the data?

Select one.

- ☐ Model A
- ☐ Model B
- ☐ Model C
- ☐ Model D
- ☐ Model E lowest BIC

Decision Trees

In the evolving domain of marine ecosystem monitoring, real-time fish species classification from underwater drone footage plays a critical role in ocean conservation efforts. This technology aids in identifying fish species to track biodiversity and monitor endangered populations.

The data collected comes from individual video frames captured by underwater drones equipped with advanced sensors. Several important features are extracted from these frames, which serve as input to a decision tree classifier for real-time species identification. Among these features, three are especially critical:

- **Hue-Saturation-Value (HSV)**: A continuous variable representing the color distribution within the frame. HSV is particularly useful in distinguishing fish species based on unique coloration patterns visible in their natural habitat.
- **Water Temperature (WT)**: Another continuous variable, recorded in degrees Celsius, as different species tend to prefer specific temperature ranges. WT helps refine classification by correlating species occurrences with water conditions.
- **Behavioral Pattern (BP)**: A categorical variable describing the activity state of the fish. This feature takes values like 'feeding', 'migrating', or 'resting', providing valuable information for distinguishing between species with similar appearance but different behavioral tendencies.

Using these features, a decision tree classifier divides the input space, enabling rapid species classification for conservationists and marine biologists. This classification empowers data-driven decisions, such as identifying vulnerable populations or environmental changes. See Information Gain Resources for more details.

$$H(BP) = -\sum P(x) \cdot \log_2 P(x)$$

$$= -\left(\frac{3}{8}\log_2\frac{3}{8} + \frac{3}{8}\log_2\frac{3}{8} + \frac{1}{4}\log_2\frac{1}{4}\right)$$

$$= -(-0.531 - 0.531 - 0.5) = 1.5625$$

The dataset for classifying fish species from underwater drone footage can be seen in Table 6 below.

ID	HSV (Color Range)	Water Temperature (°C)	Behavioral Pattern (BP)	Fish Species
1	0.75	24.0	feeding	clownfish
2	0.62	28.5	migrating	tuna
3	0.83	19.0	resting	angelfish
4	0.70	26.0	feeding	clownfish
5	0.58	27.5	migrating	tuna
6	0.85	20.0	resting	angelfish
7	0.77	25.0	feeding	clownfish
8	0.61	29.0	migrating	tuna

feed = 3
mi - 3
rest = 2

Table 6: Fish Classification Dataset

This innovative application highlights the potential of decision trees in ecological research, promoting sustainable ocean resource management by accurately identifying fish species in varying environmental contexts.

7.10 Machine Learning Q10 [2pts]

Given the possible behavioral patterns (feeding, migrating, and resting), calculate the entropy of the Behavioral Pattern feature, $H(BP)$, to determine the uncertainty in the distribution of fish activities in the dataset. Round your answer to 3 decimal places.

Answer as a decimal rounded to 3 decimal places. 1.563

7.11 Machine Learning Q11 [2pts]

We are deciding between two split options for the first node in the decision tree (splitting all of the data). The two potential conditions for splitting our decision tree are as follows:

1. Split A: Split based on the condition HSV > 0.70. (1,3,6,7)
2. Split B: Split based on the condition Water Temperature < 25.0°C.

Which split should we use?

Select one.

- [x] Split A, because it has the higher Information Gain and is therefore more useful.
- [ ] Split B, because it has the higher Information Gain and is therefore more useful.
- [ ] Either split because split A and split B have equal Information Gain and are therefore equivalently useful.
- [ ] Cannot decide due to lack of information.

$HSV \leq 0.7 = (2,4,5,8)$

$H(A) = -(0.5\log_2 0.5 + 0.5\log_2 0.5) = 1$

$H(B) = -(0.75\log_2 \frac{3}{4} + 0.25\log_2 0.25)$

$= -(0.75 \times (-0.415) + 0.25(-2)) = 0.811$

$(HA)_{wt} = \frac{4}{8} \times 1 + \frac{4}{8} \times 0.811 = 0.9055$

$IG = 1.563 - 0.9055 = 0.6575$ ✓

WT < 250 {1, 3, 6, 7} feed = 1
≥ 250 {2, 4, 5, 8} rest = 2
P(f) = 1/3, P(r) = 2/3

$H(<25) = -\left(\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right)$

$= -\left(\frac{1}{3} \times (-1.585) + \frac{2}{3} \times (-0.585)\right) = 0.918$

$H(>25) \Rightarrow$ $P(m) = 3/5 = 0.6$

$P(f) = \frac{2}{5} = 0.4$

$H(>25) = -(0.6\log_2 0.6 + 0.4\log_2 0.4)$

$= -(0.6 \times (-0.737) + 0.4 \times (-1.322)$

$= 0.971$

$I(>25)_{wt} = \frac{3}{8} \times 0.918 + \frac{5}{8} \times 0.971 = 0.951$

$IG = 1.563 - 0.951 = 0.612$

Pattern Recognition Through Time Q1 [3pts]

$song_b$ \ $song_a$	21	23	22	25	26	22	19	16
17	26	28	**29**	33	38	**39**	24	20
15	22	**24**	25	29	**34**	35	22	19
20	**16**	18	**19**	23	28	**29**	18	**18**
18	15	17	18	**22**	**B** 27	27	**17**	**C** 14
15	**12**	**14**	15	19	23	**23**	16	**12**
17	6	8	**9**	12	**16**	16	12	11
20	2	**4**	4	**7**	11	11	**10**	14
22	**1**	2	2	5	**9**	9	12 **A**	**18**

Figure 22: Grid for $song_a$ vs. $song_b$

8.1 Pattern Recognition Through Time Q1a [0.5pts]

What is the value of **A** in Figure 22 above ($song_a$ vs. $song_b$)?

Answer as a whole number. 12

8.2 Pattern Recognition Through Time Q1b [0.5pts]

What is the value of **B** in Figure 22 above ($song_a$ vs. $song_b$)?

Answer as a whole number. 27

8.3 Pattern Recognition Through Time Q1c [0.5pts]

What is the value of **C** in Figure 22 above ($song_a$ vs. $song_b$)?

Answer as a whole number. 14

8.4 Pattern Recognition Through Time Q1d [0.5pts]

What is the distance between $song_a$ and $song_b$ based on Figure 22?

Answer as a whole number. distance between Songs: 20

8.5 Pattern Recognition Through Time Q1e [1pt]

What is the path in Figure 22? 1 -2-2-5-9-9-10-12-1617-18 - 19-20

Answer as a path of numbers (fill in the blanks).

Pattern Recognition Through Time Q2 [3pts]

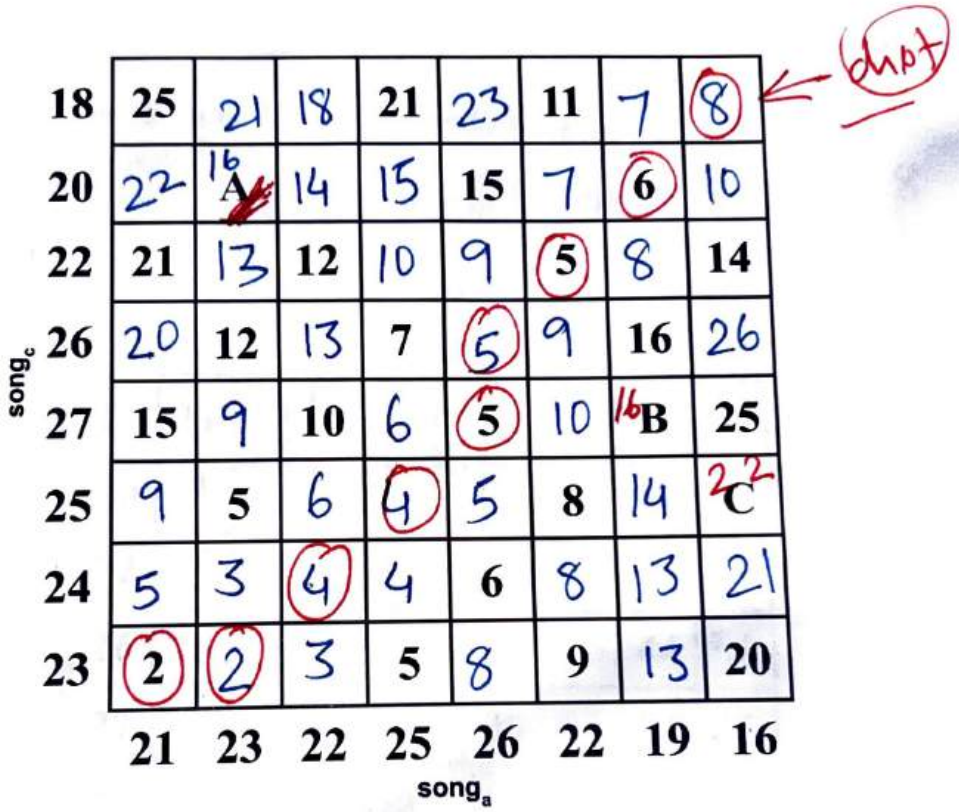


Figure 23: Grid for $song_a$ vs. $song_c$

$0^2+2^2+0^2+1^2+0^2+0^2+1^2+2^2$

$\sqrt{10}$

8.6 Pattern Recognition Through Time Q2a [0.5pts]

What is the value of **A** in Figure 23 above ($song_a$ vs. $song_c$)?

Answer as a whole number. A = 16

8.7 Pattern Recognition Through Time Q2b [0.5pts]

What is the value of **B** in Figure 23 above ($song_a$ vs. $song_c$)?

Answer as a whole number. B = 16

8.8 Pattern Recognition Through Time Q2c [0.5pts]

What is the value of **C** in Figure 23 above ($song_a$ vs. $song_c$)?

Answer as a whole number. C = 22

8.9 Pattern Recognition Through Time Q2d [0.5pts]

What is the distance between $song_a$ and $song_c$ based on Figure 23?

Answer as a whole number. 8

8.10 Pattern Recognition Through Time Q2e [1pt]

What is the path in Figure 23?

Answer as a path of numbers (fill in the blanks). 2-2-4-4-5-5-5-6-8

Pattern Recognition Through Time Q3 [3pts]

$song_c$ \ $song_b$	22	20	17	15	18	20	15	17
18	21	17	16	19	19	21	24	25
20	17	15	18	23	25	25	30	33
22	15	17	22	29	33	35	42	47
26	15	17	22	29	B	36	44	50
27	11	13	18	25	30	33	41	47
25	6	8	13	21	C	29	37	43
24	3	5	11	19	24	27	35	41
23	A	4	10	18	23	26	34	40

Figure 24: Grid for $song_b$ vs. $song_c$

8.11 Pattern Recognition Through Time Q3a [0.5pts]

What is the value of **A** in Figure 24 above ($song_b$ vs. $song_c$)? A = 1

Answer as a whole number.

8.12 Pattern Recognition Through Time Q3b [0.5pts]

What is the value of **B** in Figure 24 above ($song_b$ vs. $song_c$)?

Answer as a whole number. B = 33

8.13 Pattern Recognition Through Time Q3c [0.5pts]

What is the value of **C** in Figure 24 above ($song_b$ vs. $song_c$)?

Answer as a whole number. C = 26

8.14 Pattern Recognition Through Time Q3d [0.5pts]

What is the distance between $song_b$ and $song_c$ based on Figure 24?

Answer as a whole number. ~~26~~ ~~33~~ 25

8.15 Pattern Recognition Through Time Q3e [1pt]

What is the path in Figure 24?

Answer as a path of numbers (fill in the blanks). 1-3-6-11-15-15-15-16-19-19-~~21~~ 21 ↓ 24 ↓ 25

8.16 Pattern Recognition Through Time Q4 [1pt]

Professor Oak feels two of these birds are related to each other, while the third is unrelated. He believes we can identify the outlier using the songs. Can you identify which one is the outlier?

Select one.

- ☐ $song_a$
- ☐ ~~$song_b$~~
- ☐ $song_c$

a - b = ~~12~~ 20
a - c = ~~16~~ 8
b - c = ~~40~~ 25

ac dist small

Hidden Markov Models (HMMs)

Professor Oak finds out that the outlier bird lets out a specific call when a mountain lion is nearby. Upon further research, he realizes the lion belongs to a species which was considered to be extinct in the area! The professor wants to study the lion's hunting patterns in greater detail. However it is very elusive, evading cameras and humans and only hunting on specific days. Professor Oak arrives at the conclusion that the only way to predict whether the lion is hunting on a particular day or not is by listening to a specific call of the bird, which he refers to as call C.

Professor Oak collects recordings over several weeks and identifies the days when the bird lets out call C, indicating when the lion is out to hunt. Let us denote the occurrence of a call on a particular day by C and its absence by NC. Let us denote the state of the lion being out for hunting by L and its absence by NL.

The professor observes the following sequences of calls spanning over 5 days:

No call, No call, Call, Call, No call

Professor Oak wants to determine on which days the lion was hunting using the above sequence. Professor Oak has studied HMMs, but realizes that computing the probabilities of every possible sequence will be very time consuming. Confused, he visits Professor Juniper for a better approach. She recommends the Viterbi algorithm to him. The Viterbi algorithm uses dynamic programming to compute the most likely sequence, saving a lot of time. Your task is to help Professor Oak find out the most likely hunting sequence using the recording sequence with the help of the Viterbi algorithm.

On the first day, the probability that the lion is out to hunt is 0.5 and the probability that the lion is not out to hunt is also 0.5. The initial state probabilities are $\pi_L = 0.5$, and $\pi_{NL} = 0.5$.

If the lion has hunted on a given day, the probability it will hunt the next day is 0.6 and the probability that it will not hunt the next day is 0.4. If the lion has not hunted on a given day, the probability it will hunt the next day is 0.3 and the probability it will not hunt the next day is 0.7.

The transition probabilities are then given by Table 7:

		Next State	
		L	NL
Current State	L	0.6	0.4
	NL	0.3	0.7

Table 7: Transition Probabilities

$P(L \to L) = 0.6$
$P(NL \to L) = 0.3$
$P(L \to NL) = 0.4$
$P(NL \to NL) = 0.7$

If the lion is out to hunt, the probability that it will be observed that the bird lets out the call (C) is 0.85 while the probability that it will be observed that the bird does not let out the call (NC) is 0.15. If the lion is not out to hunt, the probability that it will be observed that the bird lets out the call (C) is 0.15 while the probability that it will be observed that the bird does not let out the call (NC) is 0.85.

The emission probabilities are then given by Table 8:

		Emission Probability	
		L	NL
Observed	C	0.85	0.15
	NC	0.15	0.85

Table 8: Emission Probabilities

To help with your calculations we provide a partially filled trellis in Table 9 below.

	Day 1	Day 2	Day 3	Day 4	Day 5
	NC	NC	C	C	NC
L	A	0.019125	C	D	0.0029597754375
NL	0.075	B	0.026551875	0.0038689875	0.011181373875

Table 9: Partially filled Trellis

8.17 Pattern Recognition Through Time Q5 [1pt]

What is the value of A in Table 9? Do NOT round.

Answer as a decimal.

8.18 Pattern Recognition Through Time Q6 [1pt]

What is the value of B in Table 9? Do NOT round.

Answer as a decimal.

8.19 Pattern Recognition Through Time Q7 [1pt]

What is the value of C in Table 9? Do NOT round.

Answer as a decimal.

8.20 Pattern Recognition Through Time Q8 [1pt]

What is the value of D in Table 9? Do NOT round.

Answer as a decimal.

8.21 Pattern Recognition Through Time Q9 [1pt]

What is the most likely sequence of the lions hunting patterns given the observed bird calling sequence?

Answer as a sequence (fill in the blank).

9 Logic and Planning [15pts]

Part 1

Terminus is an avid space travel=ler in the year 2424. They want to select a spaceship that can handle multiple roles. The ship they are looking for should have some combination of the following 4 features.

- Large Cargo Space (CS): The ship has a large storage capacity, enabling it to carry significant cargo or supplies for long missions or to transport goods.
- Advanced Weaponry (AW): The ship is equipped with advanced weapons and defense systems, making it suitable for high-risk missions or exploring unknown space.
- Good Living Quarters (LQ): The ship offers spacious, well-equipped living quarters. Suitable for extended journeys.
- High Maintainability (HM): The ship has accessible components and allows efficient in-flight repairs.

Terminus uses a logical circuit, shown in Figure 25 to determine what combinations of features is desirable. Your task is to analyze this circuit and construct a corresponding truth table.

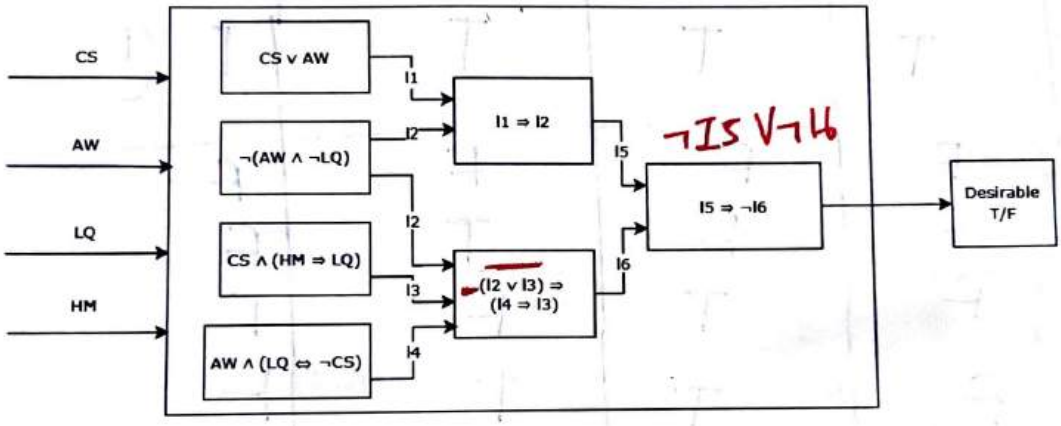


Figure 25: Logical circuit

CS ∨ AW = l1

¬(AW ∧ ¬LQ) = l2

(CS) ∧ (HM ⇒ LQ) = l3

AW ∧ (LQ ⇔ ¬(CS)) = l4

HM ⇒ LQ → ¬HM ∨ LQ

l1 ⇒ l2 = I5

I5 = l2 ∨ l3

I6 = l4 ∧ l3

desirable = (not I5) or (not I6)

I6 = (I2 or I3) <= (I4 ≤ I3)

CS	AW	LQ	HM	Desirable	
F	F	F	F	F	
F	F	F	T	F	
F	F	T	F	F	
F	F	T	T	F	
F	T	F	F	T	(B)
F	T	F	T	T	(C)
F	T	T	F	T	
F	T	T	T	T	(D)
T	F	F	F	F	
T	F	F	T	F	
T	F	T	F	F	
T	F	T	T	F	
T	T	F	F	T	
T	T	F	T	T	
T	T	T	F	F	
T	T	T	T	F	

9.1 Logic and Planning Q1 [4pts]

Given the logical circuit, select all values that are True for **Desirable** for each of the combinations in the partial truth table shown in Table 10 below. Select all that apply.

CS	AW	LQ	HM	Desirable
0	0	0	1	A
0	1	0	0	B
0	1	0	1	C
0	1	1	1	D
1	0	0	0	E
1	0	1	0	F
1	1	1	0	G
1	1	1	1	H

Table 10: Partial Truth Table

Multiple select.

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E
- ☐ F
- ☐ G
- ☐ H

9.2 Logic and Planning Q2 [2pts]

Terminus is at a spacecraft dealership. They are presented with 2 ships.

- *The Celestial* is a versatile vessel equipped with a large cargo space for hauling extensive supplies across long-distance missions and a high-tech advanced weaponry system for defensive maneuvers when encountering piracy. It lacks a comfortable living quarters, but its high maintainability is hard to miss, as it ensures easy in-flight repairs and upgrades.
- *The Voyager* is designed for space touring, with well-appointed living quarters. However, cargo space is limited and it has outdated weapons. The ship is highly maintainable, but without substantial cargo space or defenses, it may fall short on demanding missions.

According to the truth table, which ship is desirable for Terminus?

Select one.

- ☐ Only the Celestial is desirable
- ☐ Only the Voyager is desirable
- ☐ Both the Celestial and the Voyager are desirable
- ☐ Neither the Celestial nor the Voyager are desirable

9.3 Logic and Planning Q3 [2pts]

Which of the following clauses represent the initial state of the setup?

Select one.

- ☐ $Cargo(C)$
- ☐ $Cargo(C) \wedge Fuel(F)$
- ☐ $Cargo(C) \wedge Fuel(F) \wedge At(P)$
- ☐ $Cargo(C) \wedge \neg HasCargo(C) \wedge Fuel(F) \wedge At(P)$
- ☐ None of the above.

9.4 Logic and Planning Q4 [3pts]

Which of the following predicates **must** be in the clause that represent the goal state? Select all that apply.

Multiple select.

- ☐ $\neg Cargo(C_1)$
- ☐ $HasCargo(C_2)$
- ☐ $Fuel(L)$
- ☐ $\neg HasCargo(C_3)$
- ☐ $At(P)$
- ☐ $Center(C_4)$
- ☐ $\neg Fuel(F)$

Goal - cargo delivered.
$\neg Cargo(C_1) \ldots \neg Cargo(C_5)$
No cargo @ ship.
$\neg HasCargo(C)$
$Center(C)$ ship location doesn't matter.
fuel - not needed.

9.5 Logic and Planning Q5 [4pts]

Suppose the ship is currently at F_3. It has delivered cargo C_1, C_5, and has cargo C_2 on board. Cargo C_3, C_4 are still waiting for pick up at F_3, F_4, and the ship's fuel level is low.

If we perform one step of progression search, which of the following are valid next actions? Select all that apply.

Multiple select.

- ☐ $Offload(C_3, F_3)$ no C_3 on board
- ☐ $Move(F_3, F_1)$
- ☐ $Move(F_3, P)$
- ☐ $Move(F_3, F_4)$
- ☐ $Refuel(F_3)$ Refuel only at P
- ☐ $Gather(C_3, F_3)$
- ☐ $Offload(C_2, F_3)$ not allowed

Curr State - $At(F3)$, $HasCargo(C_2)$, $Fuel(L)$, C_3 & C_4 @ F3 & F4

10 Planning Under Uncertainty [15pts]

Introduction

Leela operates an interplanetary delivery service operating between Earth and Mars. Leela's delivery spaceship has limited cargo capacity and needs to decide which packages to accept for transport to maximize profit.

There are three types of packages:

1. Small, high-priority packages with a fixed profit of $3,000 each. They take up 1 unit of cargo space.
2. Medium-sized packages with variable profit. There's a 60% chance of earning $8,000 and a 40% chance of earning $4,000 per package. They take up 2 units of cargo space.
3. Large, mysterious packages. The profit is unknown but rumored to be very high or potentially a loss. They take up 3 units of cargo space.

Leela's spacecraft has a cargo capacity of 5 units. Leela has 3 decision points representing opportunities to accept or reject packages before the spacecraft departs for Earth. At each decision point, Leela can receive a request for one package of any type. Leela would like to maximize the total expected profit from delivered packages.

10.1 Planning Under Uncertainty Q1 [1pt]

What is included in the state space? Select all that apply.

Multiple select.

- ☑ Remaining decision points
- ☐ Probability of accepting Package A
- ☐ Probability of accepting Package B
- ☑ Available cargo space

A = Small
B = Med
C = L

10.2 Planning Under Uncertainty Q2 [1pt]

What is included in the action space? Select all that apply.

Multiple select.

- ☑ Accept Package (if enough cargo space)
- ☑ Reject Package
- ☒ Transition probabilities to next state
- ☒ Available cargo space State space

10.3 Planning Under Uncertainty Q3 [1pt]

What is the reward for accepting Package A? Small = $3K

Answer as a whole number. 3000

10.4 Planning Under Uncertainty Q4 [1pt]

What is the reward for accepting Package B?

Answer as a whole number. 6400

$8000 \times \frac{60}{100} + 4000 \times \frac{40}{100}$

expected = 4800 + 1600
= 6400

10.5 Planning Under Uncertainty Q5 [1pt]

Which of the following statements are true if we only consider packages A and B (assuming package C is not an option)? Select all that apply.

Multiple select.

- [x] A terminal state is reached when no decision points remain.
- [x] A terminal state is reached when no cargo space remains.

10.6 Planning Under Uncertainty Q6 [1pt]

Suppose Leela's cargo consist of 2 medium packages and 1 small package. What is Leela's expected profit?

Answer as a whole number.

10.7 Planning Under Uncertainty Q7 [1pt]

Is the policy of carrying 2 medium packages and 1 small package optimal?

Select one.

- [x] Yes
- [ ] No

10.8 Planning Under Uncertainty Q8 [2pts]

Which of the following statements are true? Select all that apply.

10.5 Planning Under Uncertainty Q5 [1pt]

Which of the following statements are true if we only consider packages A and B (assuming package C is not an option)? Select all that apply.

Multiple select.

- ☑ A terminal state is reached when no decision points remain.
- ☑ A terminal state is reached when no cargo space remains.

10.6 Planning Under Uncertainty Q6 [1pt]

Suppose Leela's cargo consist of 2 medium packages and 1 small package. What is Leela's expected profit?

Answer as a whole number.

10.7 Planning Under Uncertainty Q7 [1pt]

Is the policy of carrying 2 medium packages and 1 small package optimal?

Select one.

- ☑ Yes
- ☐ No

10.8 Planning Under Uncertainty Q8 [2pts]

Which of the following statements are true? Select all that apply.

Multiple select.

- ☑ Accepting Package C allows Leela to learn about its profit potential.
- ☐ Rejecting Package C allows Leela to learn about its profit potential.
- ☐ Accepting Package C allows Leela to maximize short term gain but potentially miss out on long term gain.
- ☑ Rejecting Package C lets Leela stick to known profits of packages A and B, maximizing short term gain but potentially missing out on long term gain.

10.9 Planning Under Uncertainty Q9 [3pts]

What methods can be applied to assist when deciding whether to pick package C? Select all that apply.

Multiple select.

- ☑ Thompson sampling
- ☑ UCB (Upper confidence bound) heuristic
- ☑ Gittins Index
- ☐ Dynamic Time Warping

11 Machine Learning - Extra Credit [2pts]

All questions in this section are extra credit.

Advancing Agricultural Research with Convolutional Neural Networks (CNN) for Orange Variety Identification

Dr. Muzan has launched a groundbreaking project on classifying orange varieties to support agricultural practices and improve crop quality. The dataset consists of 50,000 high-resolution images of six distinct orange varieties: Valencia, Mandarin, Dream Navel, Kumquat, Torocco, and Chinotto as shown in Figure 26. These varieties exhibit subtle differences in size, color, texture, shape, and rind patterns, which makes precise classification challenging. Accurate identification is essential for inventory management, crop quality monitoring, and market optimization.

For theoretical insights, Dr. Muzan refers to Chapter 21 of Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig, which discusses neural networks and their applications in detail.

Initially, Dr. Muzan opted for a relatively straightforward approach by designing a multi-layer perceptron (MLP) with 6 outputs. His MLP model includes:

Input Image (64 x 64 x 1) $= 64 \times 64 = 4096$ px

↓

Linear Layer 1 (425 hidden units, bias term, activation function: ReLu) $4096 \times 425 + 425$

↓

Linear Layer 2 (255 hidden units, bias term, activation function: ReLu) $425 \times 255 + 255$

↓

Linear Layer 3 (80 hidden units, bias term, activation function: sigmoid) $255 \times 80 + 80$

↓

Output Layer (6 units, bias term, activation function: SoftMax)

↓

output (6 probabilities)

IP → Linear : 1742825

L1 → L2 : 108630

L2 → L3 : 20480

L3 → OP : $80 \times 6 + 6 = 486$

1872421

connected layers than in our original MLP model. Dr. Muzan then designed a Convolution Neural Network for the task at hand with the architecture below.

- CONV-K-N = convolutional layer with N filters, each size K×K.
- Padding and stride params are consistently set to 0 and 1, respectively.
- POOL-K = K × K pooling layer with a stride K and zero padding.

Recall that there is one bias parameter is added for each filter. For example, in the first convolutional layer (CONV-5-32): there are 32 biases (one per filter).

Input Image (128 x 128 x 1)
↓
Convolution layer(2D) (CONV-5-32)
↓
Max Pooling (2D) (POOL-2)
↓
Convolution layer(2D) (CONV-3-16)
↓
Max Pooling (2D) (POOL-2)
↓
Linear Layer 3 (50 hidden units, bias term, activation function: sigmoid)
↓
Output Layer (6 units, bias term, activation function: SoftMax)
↓
output (6 probabilities)

To compare the parameters from a basic feed-forward network and a Convolution Neural Network, Dr. Muzan would like to know how many learnable parameters there are in the above CNN architecture.

Answer as a whole number.

11.4 Machine Learning Extra Credit Q4 [0.4pts]

Dr. Muzan is testing different optimization algorithms to improve the training efficiency of his CNN. During experiments, he notices that with an initial learning rate of 0.01, one optimizer adapts its learning rate over time, resulting in faster convergence compared to a fixed-rate optimizer. Which of the following optimizers is known for having this adaptive learning rate capability?

Select one.

- ☐ Stochastic Gradient Descent (SGD)
- ☐ Adam Optimizer
- ☐ Momentum Optimizer
- ☐ Nesterov Accelerated Gradient (NAG)

2nd Conv (2D) filter = 16
fsize = 3×3
stride = 1, Padding = 0

total wt = 16 × 3 × 3 = 144
total bias = 16 × 1 = 16
params = 144 + 16 = 160

O/p size = $\left(\frac{62-3}{1}+1\right) = 60$

pool-2 — $\left(\frac{60}{2}\right) = 30$

O/p = 30 × 30 × 16

Fully connected — 3 lr, 50 Hidden.

I/p = 30 × 30 × 16 = 14400
wt = 14400 × 50 = 720,000 wt.
biase = 50
total params = 720,050

O/p layer = 6 units.
I/p = 50
wt = 50 × 6 = 300 + 6 bias = 306 params

Total = 721348